Vineet Vora

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Summary

Engineer who takes AI systems from an ambiguous problem to something deployed and used. MS in Artificial Intelligence at NJIT. Built a natural-language-to-SQL system letting non-technical users query unfamiliar databases in plain English, on open-source models only so it runs inside a customer's own environment. Comfortable owning the whole line — scoping, building, and explaining the result to people who don't write code.

Technical Skills

Languages: Python, Java, TypeScript, JavaScript (ES6+), SQL
AI / ML: PyTorch, HuggingFace Transformers, vLLM, LoRA / PEFT, TensorFlow, Scikit-learn, OpenCV
Data & Infrastructure: Hadoop, MapReduce, HDFS, YARN, AWS EC2, SLURM, Docker, CUDA, PostgreSQL, MongoDB
Web & Delivery: React.js, Node.js, Cloudflare Pages, Git, CI/CD, scoping, Design Thinking (IBM)

Selected Work

Natural Language → SQL over Enterprise Databases (MS Thesis) *Jan 2026 – Present*
Python, vLLM, PyTorch, Qwen3-Coder-30B, SLURM, SQLite

- Turns plain-English business questions into executable SQL against multi-table databases the model has never seen: schema linking, candidate generation, targeted decomposition, self-repair, and consensus voting.
- Reached **70.2% execution accuracy on BIRD** (1,534 questions, 11 real-world databases) using **only open-source models and no proprietary API** — so it runs fully inside a customer's own environment with no data leaving it.
- Served 30B-parameter models on a 2×A100 SLURM cluster with vLLM; built checkpoint/resume, timeouts and graceful degradation so an 8-hour evaluation never restarts from zero.
- Compared four open backbones to isolate what moved accuracy, then cut five candidate-selection variants to one after measuring the extra four bought nothing.

Geomagnetic Storm Forecasting from NASA Solar Imagery *Dec 2025 – Present*
github.com/VineetV2/imf-bz-forecasting-surya | Python, PyTorch, LoRA, CUDA, Weights & Biases

- Adapted Surya, a 366M-parameter Vision Transformer pretrained on NASA SDO imagery, to forecast interplanetary magnetic field Bz at +24h/48h/72h — what utilities watch before a storm.
- Compared LoRA against a frozen encoder on 13-channel AIA EUV and HMI inputs (512×512); the frozen encoder reached **3.39 nT RMSE** on **0.34% of parameters**, **3.5**× more efficient than LoRA.
- Validated by Leave-One-Flare-Out cross-validation across 57 M5.0+ and X-class flares; 96.5% of folds succeeded.

Distributed Flight Analytics — 118.9M Records on Hadoop *Nov – Dec 2025*
github.com/VineetV2/hadoop-flight-data-analysis | Java, Hadoop MapReduce, AWS EC2

- Stood up a 6-node Hadoop cluster on AWS EC2 and processed 22 years of US flight data — **118.9M records, 11.66 GB** — reaching **100% data-local map execution** and **2.73× speedup** under horizontal scaling.

Peon — Local-First Memory System for AI Coding Agents *2026*
github.com/VineetV2/peon-mem | TypeScript, MCP, SQLite

- Shipped an MCP server giving AI coding agents persistent project memory: automatic capture, LLM consolidation, hybrid retrieval, daily self-audit. Drops into any MCP client.

Leadership

VITOPIA, VIT-AP University Flagship Festival — Head of Sponsorship & Core Team *Jan – Feb 2024*

- Led a **20-person team** through outreach to **450+ companies**, closing **$180,000** (INR 1.5 Cr) in corporate sponsorship and exceeding every revenue target.
- Owned the commercial relationship end to end — pitch, negotiation and partner management with non-technical stakeholders — alongside vendor, venue and scheduling logistics.

Beat The Heat, VIT-AP — Club Administrator & Head of Finance *Aug 2022 – Jan 2025*

- Ran the club's annual budget and supplier negotiations for two and a half years, setting the financial strategy the committee planned expansion against.

Publication

Preventing Wildfires in Energy Transmission by Automatic Power Line Defects Detection Using Machine Learning and AI
2025 New Jersey Future Energy Transmission Conference (NJFET), IEEE — Dec 2025 | presented | DOI 10.1109/NJFET67489.2025.11380637

- Computer vision framework moving utilities from reactive repair to preventive maintenance on power line defects and vegetation encroachment in UAV imagery; built the falling-wire proof of concept with Canny edge detection and a Probabilistic Hough Transform.

Education

New Jersey Institute of Technology — MS, Artificial Intelligence | GPA 3.83 | Newark, NJ *Expected Jun 2027*
Advanced NLP, Computer Vision, Information Retrieval, ML for Data Science, Big Data

Vellore Institute of Technology – AP — BTech, Computer Science | India *Jun 2025*